DISRUPTION INVESTOR

September 2025

Inside our AI Phase 2 ETF

Greetings from Abu Dhabi!

If you didn't know, my family and I (Stephen) packed up our lives and made the move to "the capital of capital" a few weeks ago. We're loving it so far. Hot take: 100 degrees is bearable when you have a chilled pool.

My kids are already settling into their new school beautifully.

The reason we moved here was Abu Dhabi's ambition. I often say 95% of breakthrough innovation is generated by American entrepreneurs, in America. But I genuinely believe there's no better crucible for turning that innovation into reality than here.

This matters because being pro innovation and disruption is like a giant healing balm for the country. The average Emirati is inherently optimistic because he's seen technology transform his life over the past few decades.

Air conditioning made desert life livable. Nuclear power gave The Emirates clean, safe, stable electricity. For thousands of years, people in this part of the world prayed to the rain gods. Now, cloud seeding hands us water on demand.

I want that ambition, optimism, and "can do" attitude to rub off on my kids.

Abu Dhabi is also fast becoming an artificial intelligence (AI) superpower.

In this month's issue

Key prices

S&P 500	6,387.36
1-Month (Gain) +0.90%	
Nasdaq	21,158.94
1-Month (Gain) +0.51%	
Russell 2000	2,350.83
1-Month (Gain) +6.26%	
Dollar Index	98.12
1-Month (Loss) -0.58%	
US 10-Year Yield	4.265%

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Every UAE resident got a lifetime AI subscription courtesy of the government. As an aside, I wonder how much that adds to OpenAI's revenue? Starting this year, AI was integrated into the UAE's K-12 curriculum.

Abu Dhabi is also spending hundreds of billions of dollars building AI data centers. The UAE allocated 5 gigawatts of power to this project. That's enough energy to power 4 million US homes for an entire year.

Plus, it plans to buy around half a million top-tier **Nvidia (NVDA)** GPUs per year.

As it's so easy to build things here (the skyline is dominated by cranes), I think the Gulf will house the world's largest data centers.

The view from my new office…

Longtime members know we were early to the AI boom.

We first invested in Nvidia in September 2020.

We invest through the lens of disruption because, after decades in the markets, Chris and I learned the surest way to build real wealth is to own great businesses riding megatrends.

Disruption, when you're positioned right, is the closest thing investing gives you to a cheat code.

The hard part isn't spotting the trend. It's holding on for the ride. Disruption rarely plays out in tidy bursts. It runs longer than most investors can imagine.

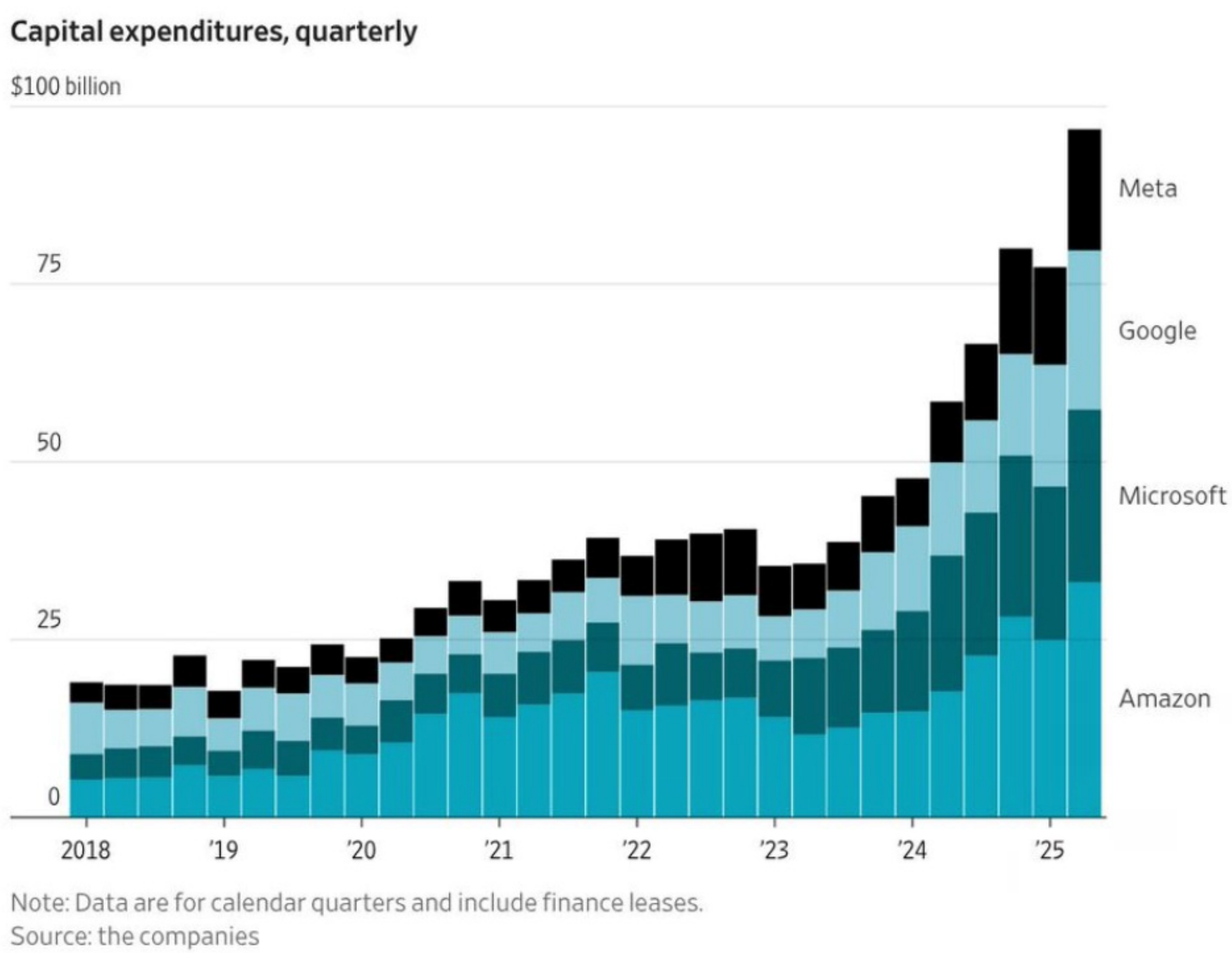
The internet wasn't a "theme," but a 30-year revolution. "Software is eating the world" was the single biggest driver of financial returns for almost 15 years.

AI is cut from the same cloth. It's a structural shift that will reshape industries and economies for the next 10–20 years. I think it could be far more disruptive than the internet because it will touch so much more of our lives.

Internet didn't really change healthcare, education, or defense—three of the biggest industries in America. But AI is already transforming these sectors, and we're only getting started.

It sounds so obvious to say, "Invest in AI." We were early to this trend. But when I step back and look at what's driving not just markets, but the economy and geopolitics today, AI tops the list.

US big tech companies will spend over $300 billion building out their AI data centers this year alone. The AI buildout is so big, it's acting as a kind of private stimulus project for the US economy.



Source: The Wall Street Journal

Our research shows total dollars spent on "compute" will exceed spending on oil in the next decade. AI tokens are replacing oil as the world's most valuable commodity.

And now, we're entering the second phase of the AI buildout.

Phase 1 was all about America's big tech giants spending like drunken sailors, scooping up as many Nvidia GPUs as they could to train the biggest models possible.

That alone created a $300+ billion annual spending surge. Big tech will spend more on AI this year than Switzerland spends running its government.

We thank them for the profits they've handed us.

But here's the question I asked myself this month: *Where will the next trillion dollars of AI spending come from, and who profits?*

Phase 2 is about three massive shifts now underway that will define the largest infrastructure buildout in history.

Each one funnels money toward a handful of companies set to rake in tens of billions of dollars. Let's review each shift below…

The 3 big shifts rewiring AI

Shift #1: Sovereign AI: A brand new, deep pocketed customer

People often compare today's AI boom to the dot-com bubble, with Nvidia as the **Cisco Systems (CSCO)** of its day.

The big difference is that Cisco's customers were debt-fueled dot-com startups with no revenue. Nvidia's customers are the *largest, most profitable companies in the world*.

As mentioned, US big tech companies will spend over $300 billion building out their AI data centers this year. That was Phase 1.

Phase 2 is even bigger: Sovereign AI.

Sovereign AI means every country is building its own "AI factory." AI models trained on local data, running on local compute, under local control.

If you believe, as I do, that AI will run healthcare systems, defense planning, and education in the coming decades… do you really want those systems sitting on foreign chips in foreign data centers?

Imagine running America's nuclear fleet on Chinese AI. Exactly. It makes TikTok's "spy app" look cute.

Governments are already writing billion-dollar checks to make sovereign AI possible:

- **Europe** is building out national supercomputers to give startups access to AI compute.

- **The UK** just switched on its most powerful supercomputer ever.

- **India** approved a 38,000 GPU national cluster to train models in 22 local languages.

- **Saudi Arabia** earmarked $100 billion to turn itself into a regional AI powerhouse.

Canada, Japan, and South Korea are all in, too.

Mark my words: Soon, governments will spend more money on AI than big tech.

Nvidia's Jensen Huang put it best: *"Every country will have an AI factory, just like every country has a telco."* This is effectively a giant wealth transfer from the rest of the world to US AI companies that make the AI gear. And that's a great thing for clued-in investors.

<u>A few things I've learned about the UAE's AI strategy before we move on.</u>

Just as barrels of oil powered the 20th century, the UAE sees compute as the most valuable resource of the 21st century. Yesterday's wealth came from pumping oil out of the ground. Tomorrow's will come from spitting out intelligence on demand.

The Gulf, led by the UAE and Saudi, has four huge advantages when it comes to AI:

1. Cheap energy.

2. Geography (within 100-millisecond latency of 4 billion people).

3. Easy to build (a place where megaprojects are possible).

4. Piles of cash.

That's why the Gulf is on its way to becoming an AI superpower. Goodbye, petrodollars—hello, PetroAI.

Sovereign AI is a global stimulus plan in disguise. I remember my friend Louis Gave—who runs the billion-dollar firm Gavekal—saying, "A winning strategy in China has been to buy whatever Xi Jinping invests in."

The same logic applies here. Politicians are telling us exactly where they're going to pour hundreds of billions of dollars. Betting alongside them is one of the easiest trades in markets today.

Shift #2: From training to inference: compute goes prime time

When people talk about AI infrastructure, they usually mean thousands of GPUs crunching away for weeks to train a giant model. That's the "training" phase.

Training is like building the engine of a Ferrari. You do it a handful of times, and it's costly and complex. But once it's built, it's ready to race.

The real money is in driving the Ferrari every day, aka "inference." Every time you ask ChatGPT a question or have it generate an image, that's inference. And it's exploding because more and more people are using AI.

Last spring, **Google's (GOOGL)** Gemini models were processing 480 trillion tokens per month. That number has doubled to 980 trillion. ChatGPT will soon hit 1 billion monthly active users!

This changes everything about AI.

Training can be done in a remote desert data center using cheap power. Inference needs to sit closer to the user, run 24/7, and deliver answers in real time.

If you thought training AI was costly… oh boy, wait until you see what we need now. Inference compute now represents most of the cost of running advanced AI models.

Nvidia isn't the big winner here. Walk into a data center, and alongside the GPUs you'll see giant cooling fans… storage disks… memory chips… networking cables… and so on.

More "inference" means more heat… more energy consumed… more memory needed… and more data flowing through thick networking cables.

This shift changes *where the money flows*.

To move data faster between chips, these clusters need more connections per rack and faster optical links, driving a massive upgrade cycle in optics.

On Nvidia's new GB200 racks, the optics bill alone can top $500,000. And demand for low-latency communication favors cutting-edge AI networking solutions like Ethernet.

The key takeaway is we're shifting from huge, one-off training runs to continuous infrastructure spend. Inference creates sustained demand for AI gear. If the training era was Nvidia's show, the inference era is where the broader ecosystem shines.

Get ready for a mad dash to overhaul AI data centers.

Shift #3: Data center spending is rotating to power, cooling, and networking

In 2006, Google built its first serious data center in Oregon. It cost about $600 million, which was a big deal at the time.

Fast-forward to 2025, and the bill for OpenAI's "Stargate" project with **Oracle Corp. (ORCL)** and SoftBank is coming in at $500 billion! Individual sites within this project will consume more power than a small city.

These aren't your father's data centers. One AI rack now guzzles the same amount of electricity as a dozen legacy racks. **Meta Platforms (META)** even tore down a half-built campus in Utah to rebuild it from scratch for AI workloads.

The big change is that AI data centers must act as one giant computer, not a warehouse of single servers.

Training and inference require every GPU to talk to every other GPU at blistering speeds. That's why companies are packing chips closer together and spending billions on optical networking gear.

The bottleneck is no longer chips. It's power. A rack of servers stuffed with AI chips needs about 10X more power than a normal "cloud" server. Meta's Hyperion cluster in Louisiana alone will consume roughly as much power as all of San Francisco.

Electricity = oil for AI.

AI's huge power demands are also forcing the whole industry to shift to liquid cooling.

Old-school cloud centers were cooled like office buildings with giant air conditioners pushing cold air through rows of servers. But AI chips run so hot, air can't carry the heat away fast enough to keep them from melting (literally).

That's why liquid cooling is becoming to go-to. Water is 17X more effective than air at removing heat.

Google and **Microsoft (MSFT)** are ripping out old air-cooled systems and retrofitting water loops right up against the chips. At Google's Oklahoma campus, pipes literally snake through the racks, pumping chilled, treated water directly to the GPUs.

The easiest way to picture the AI buildout is to break data centers into three buckets of spending:

1. **The computers (~60% of the bill):** This is the heart of the machine: GPUs, servers, racks, networking gear, and fiber optics. A few years ago, chips alone were 80% of the tab. Now, it's closer to 50% as networking and optics balloon in importance.

2. **Power and cooling (~25% of the bill):** Think of this as the "life support system." It includes transformers, switchgear, and liquid-cooling plants. This used to be barely 10% of the total cost. Now it's the fastest-growing expense line. GPUs run so hot that air can't cool them anymore.

3. **The site (~15% of the bill):** Finally, there's the real estate, land, and building shells. Major line items in this category include long-term power contracts, grid upgrades, and on-site gas turbines.

The next trillion dollars of AI spend won't be on chips alone…

GPUs made Phase 1 billionaires. The pipes and power will mint Phase 2 fortunes.

We're early, again.

I find it ironic the sexiest technology on Earth—artificial intelligence—depends on boring inputs like electricity, thick cables, and cooling towers. That's where the puck is going.

I'll hand you off to Chris to show you the three disruptors we're adding to our AI Phase 2 "ETF"… and how they're best-positioned to ride this wave for the next decade.

How we'll profit from Phase 2 of the AI infrastructure buildout

We're going to capitalize on Phase 2 of the AI infrastructure buildout by building our own "ETF"—like we did for cybersecurity in November 2024.

Specifically, we're going to buy the top three companies poised to benefit from the three big shifts in AI investment Stephen talked about:

- Sovereign AI
- Inference
- Data center spend rotating to networking, power, and cooling

As you probably know, an ETF is an investment fund that trades like a stock. There's no actual ETF focused on Phase 2 of the AI infrastructure buildout because it's an investment theme we came up with and defined ourselves.

Even if there were such an ETF, it would hold too many stocks for our liking—as well as too many of the wrong stocks—like every other ETF. That's why we're building our own.

Oftentimes, we prefer to simply invest in the one company we see as the best business to capitalize on a disruptive megatrend. But that's not the best approach for this theme because there will be multiple winners providing different types of complementary critical infrastructure throughout AI data centers.

Note: We're only allocating one-third of a full position to each of the three stocks. In other words, take whatever you would normally invest in a *Disruption Investor* stock, divide that number by three, and invest that amount in each of these stocks. We're essentially treating these stocks like one equally-weighted fund.

I should also note that all three stocks we're adding today used to be in our portfolio. We stopped out of them earlier this year but still made big gains on two.

Specifically, we booked a gain of 87% on **Arista Networks (ANET)**, a gain of 74% on **Vertiv Holdings (VRT)**, and a loss of 23% on **Broadcom (AVGO)**. But we now have extremely compelling reasons to buy back into all three.

We'll place a 40% hard stop on each stock in our Phase 2 AI infrastructure ETF. This will protect us from big losses if one of the positions moves against us, which is always a possibility.

With that, let's dive into the three stocks we're recommending today and how they'll benefit from the three big shifts happening in AI.

Recommendation #1: Broadcom (AVGO)

As Stephen said, "Sovereign AI means every country is building its own 'AI factory.'"

When a country like the UK or India builds an AI data center (or "factory"), it needs to connect thousands of GPUs (and/or other AI accelerators) in a cluster, so they all work together. That requires specialized networking chips to shuttle data at extremely high speeds among the processors (brains) and servers.

Broadcom is the world's leading provider of these networking chips. They work like traffic control centers directing data, so the right information goes from point A to point B to point C, etc.—all at the blistering speeds necessary for AI workloads, without getting lost.

Big tech companies like **Amazon (AMZN)**, Alphabet, and Meta often choose Broadcom's Tomahawk series of switch chips to connect their large AI clusters, for example.

The company's latest Tomahawk 6 chips deliver the world's first 102.4 terabits per second of switching capacity. That's like being able to handle data flows equivalent to streaming 20 million high-def **Netflix (NFLX)** movies simultaneously.

This high capacity means fewer networking switch chips are needed to perform the task. But you still need a lot of these chips because of the astronomical amounts of data AI factories must move among thousands of processors.

An AI cluster of 100,000 accelerators, for example, might need 4,000 or so Tomahawk 6 switch chips, creating a massive market opportunity for Broadcom as countries build out their sovereign AI infrastructure.

Simply put, every new sovereign AI data center around the world is a potential customer for Broadcom's high-end networking chips.

Broadcom also stands to benefit handsomely from sovereign AI through its VMware subsidiary, which enables countries to build private AI clouds to keep sensitive data safe.

VMware's Cloud Foundation (VCF) platform, for example, provides the software infrastructure that governments need to maintain complete control over their AI operations. This includes data sovereignty controls ensuring information never leaves national borders, zero-trust security architecture, and compliance with relevant regulations throughout the world.

VCF is like an operating system for the whole data center. The latest version, VCF 9, has been touted as an "AI factory in a box."

Broadcom also makes a bunch of other important products for AI data centers, like network interface cards, PCI Express links, storage controllers, and optical transceivers. Together, these products comprise much of the connective tissue that allows AI factories—whether sovereign or otherwise—to function.

It's no wonder Broadcom's motto is "connecting everything."

When it comes to the shift from AI training to inference, Broadcom stands out because of its custom ASICs (application-specific integrated circuits).

We talked about this in the October 2024 issue of *Disruption Investor* when we first recommended Broadcom—before stopping out when the market crashed in early April.

The gist of the story here is that big tech companies don't want to be beholden to Nvidia and its GPUs forever. Plus, you can gain efficiencies in speed and cost over general-purpose GPUs when you build custom AI accelerators for specific sorts of workloads.

So big tech is now spending billions developing their own AI chips—custom ASICs. And they're teaming up with Broadcom to do it.

An ASIC is like a high-quality chef's knife. It's designed to be the best tool for a specific purpose.

This has big implications for AI inference.

Basically, AI training is learning, and AI inference is doing. Every time you interact with AI—like when you ask ChatGPT a question and it answers—that's inference. The training happened earlier behind the scenes before the AI was released into the wild.

Meanwhile, you only need to train an AI model once and update it periodically, whereas inference happens every time someone uses an AI-powered app or service. So we're talking about billions of times a day across countless organizations and billions of devices.

That means AI inference is a much broader—and likely a much larger—market than AI training. This bodes extremely well for Broadcom, because custom ASICs can be designed specifically to optimize AI inference.

So these custom ASICs can provide higher performance and efficiency compared to GPUs and CPUs. This can lead to faster processing times, lower power consumption, and reduced costs, making custom ASICs ideal for large-scale AI inference applications.

Broadcom is the global leader in custom ASIC chip designs with a market share of 60% to 80%, according to industry analysts.

The third major shift in Phase 2 of the AI infrastructure buildout is a direct consequence of the first two: data center architecture itself is being completely rethought for AI, and the industry is pouring money into overcoming new bottlenecks in power, cooling, and networking.

In other words, we've entered an era where the bottleneck in AI is no longer computing power itself, but everything around it—can we feed the beast enough power, cool it, and move data into and out of it fast enough?

Broadcom's high-speed networking gear is squarely focused on solving part of that bottleneck.

Meanwhile, the three shifts that define Phase 2 are already visible in Broadcom's financial results. The company's AI-related revenue is growing at a sustainable annual rate of about 60%. That means Broadcom will generate about $20 billion in AI-related revenue this year and north of $30 billion in 2026.

Action to take: Buy one-third of a full position in Broadcom (AVGO) at current market prices and set a 40% hard stop.

Recommendation #2: Arista Networks (ANET)

If Broadcom’s networking chips are like the traffic control centers for AI networking, Arista provides the actual roads and highways on which all the data travels—the switching systems, routers, and software that allow AI data centers to function.

Importantly, Arista often builds its networking hardware using Broadcom’s chips. So the solutions provided by the two companies complement each other rather than compete.

Arista is deeply entrenched with US cloud giants like Microsoft and Meta. The enormous AI clusters these companies have been building recently rely on Arista’s tech to connect tens of thousands of GPUs.

Traditionally, the biggest AI clusters used network technology called InfiniBand to connect GPUs. But InfiniBand is Nvidia’s proprietary tech.

Arista’s approach to networking uses advanced Ethernet technology—the common networking standard that’s more familiar to IT staff and capable of interconnecting anything. This is crucial for a country building a sovereign AI cloud, because they likely have a mix of equipment and want flexibility.

Arista’s Ethernet-based networking tech can seamlessly link GPUs, CPUs, custom AI chips—whatever hardware you want—to avoid vendor lock-in.

Now, as sovereign AI initiatives take off, Arista is extending its expertise to other countries. It’s partnered with Oracle, for example, to help build the UK’s AI cloud infrastructure. The company’s also recently expanded into India.

Whether it’s in London, Riyadh, or New Delhi, Arista’s high-performance switches are emerging as go-to solutions wherever a large AI data center is being built.

What’s more, the company’s CloudVision and EOS Smart AI Suite software ensure compliance and quick troubleshooting for AI administrators in sovereign settings.

Plus, its AVA (Autonomous Virtual Assist) engine acts like Sherlock Holmes for your network, using AI to answer questions you didn’t even know to ask—spotting odd patterns, warning you of potential problems early on before they become bigger issues, and sometimes even auto-fixing them.

So as AI spend broadens out to many new data centers throughout the world, Arista will supply the high-speed digital highways connecting the bulk of those AI servers.

Meanwhile, Arista’s networking tech isn’t just meant for big training clusters. It’s also designed for the needs of AI inference—balancing constant traffic, variable demand, and ultra-low latency (wait time).

Features like cluster load balancing and AI Analyzer constantly optimize data flows so inference jobs are handled quickly, even if demand spikes or patterns change.

And tools for real-time monitoring, job-level tracking, and anomaly detection make it much easier for operators to fine-tune inference performance. That's key as AI gets embedded everywhere, from customer service to healthcare and national defense in the coming years.

Finally, as AI data center spending shifts toward areas like power and networking, Arista stands out. Its high-performance, lower-power networking solutions—along with its open Ethernet ecosystem and AI-powered smart traffic tools—give customers flexibility in their infrastructure setups. This means less wasted energy and more value for every dollar spent.

Just like with Broadcom, we're already seeing these shifts start to show up in Arista's financial statements. The company's revenue jumped 30.4% year over year in the most recent quarter to $2.2 billion.

That's a huge jump for a mature networking company. Arista's CEO explicitly credited "increasing demand for AI networking solutions" as the key growth driver.

The company is even targeting $10 billion in annual revenue by 2026—two years ahead of prior plans—thanks to the AI infrastructure boom.

Action to take: Buy one-third of a full position in Arista Networks (ANET) at current market prices and set a 40% hard stop.

Recommendation #3: Vertiv Holdings (VRT)

You can't build a new AI data center—sovereign or otherwise—or add capacity to an existing one without power and thermal management technology.

And Vertiv is the top provider of this tech.

Think of an AI data center as a bustling city. A city needs a consistent and reliable supply of electricity to power homes, businesses, traffic lights, etc.

Vertiv's power management tech is like the city's power grid. It ensures electricity is available where and when it's needed in a data center. It keeps servers, cooling systems, and other critical equipment up and running.

Vertiv provides all the products necessary to manage an AI data center's power from "grid-to-chip." So, as soon as electricity enters the data center, Vertiv's equipment takes over to manage it and distribute it—in the right amounts—so everything runs correctly.

The company's power distribution units (PDUs) make sure all the different servers and other equipment get the right amount of power they need to operate efficiently.

Its DC power systems convert AC to DC power, which is necessary for many data center components.

The switchgear Vertiv provides transfers loads to backup power sources when necessary and acts like protective fuses and circuit breakers to keep equipment from overloading and short-circuiting.

And the uninterruptible power supply (UPS) systems Vertiv provides ensure the servers keep running without any hiccups to prevent data loss and service disruptions.

All these power management products work together using Vertiv's software to ensure a stable, reliable, and efficient power supply for AI data centers. So, just like a city can't function without a reliable power grid, an AI factory can't operate without the power management technology that Vertiv and companies like it provide.

What's more, Vertiv's modular solutions can be assembled and activated virtually anywhere on Earth. And they're tailored to meet whatever local energy, privacy, or security requirements a government or regulator needs—crucial for sovereign AI.

For example, the new "Colosseum" sovereign AI data center in Italy uses Vertiv's infrastructure to deliver secure, high-powered AI for Europe's public sector, including health and finance.

Meanwhile, as AI inference workloads spread across many sites, Vertiv's scalable solutions adapt to facilities of all sizes, ensuring efficient inference performance at every location.

But it's the third shift we've been talking about—data center spending rotating to things like power and cooling—that's going to move the needle most for Vertiv in the coming years.

See, back in 2020, average server rack power densities in data centers were typically in the range of 5 to 9 kilowatts (kW) per rack. Racks were loosely filled and ran warm but not dangerously hot. So big room air conditioners blowing cold air through the aisles could handle data center cooling needs.

That's no longer the case. GPUs and other AI accelerators consume much more energy than traditional CPUs. And racks are packed as tight as possible.

So, today's data centers have rack densities of 40 to more than 100 kW to meet the demands of AI, and they're climbing ever higher. Some operators are engineering for 300 to 400 kW soon. That means these racks of servers generate exponentially more heat. So much so that they'd literally melt without new cooling technologies.

That's where Vertiv's liquid-cooling tech comes in.

Liquid cooling leverages the superior thermal transfer properties of water and other fluids to cool high-density servers up to 3,000X more effectively than air.

It involves either circulating a coolant like water through cold plates in direct contact with electronic components like GPUs… or submerging electronic components in a non-conductive liquid coolant.

Vertiv has a full portfolio of liquid-cooling solutions to meet the thermal management needs of AI data centers today and tomorrow.

In the second quarter of 2025, Vertiv's revenue jumped 35% year over year to $2.64 billion. The company also reported an $8.5 billion backlog of orders—a clear sign that customers (including government-led sovereign AI projects) are racing to buy the power and cooling equipment they need in place before they can flip the "on" switch for new or expanded AI data centers.

Put simply, Vertiv is a key part of the physical backbone that enables AI data centers to operate.

Its power management systems deliver electricity to Broadcom's chips and Arista's switches, while its cooling systems remove the enormous heat generated by AI processing.

And its AI-powered monitoring software platforms integrate with both Broadcom's VMware software and Arista's network management tools to watch over everything—ensuring AI factories run at peak performance and sounding the alarm if anything needs attention.

Action to take: Buy one-third of a full position in Vertiv Holdings (VRT) at current market prices and set a 40% hard stop.

For the most up-to-date guidance on our current positions, click here to view the portfolio online.

Chris Wood and Stephen McBride